Vismay Vilas Chaudhari

585-290-2822 | vismaychaudhari7@gmail.com | linkedin.com/in/vismay-chaudhari | github.com/veeoid | vismay.vercel.app

EXPERIENCE

Software Engineer Sep 2025 - Present
Ecolab *St. Paul, MN*

- Consolidated iterative pricing reviews into one report for 2,500+ SKUs globally by owning a Python application from requirements through production and testing purchase-history-based division fallback rules.
- Replaced manual Snowflake releases across three environments with a Python deployment engine and GitHub Actions, implementing SQL validation, dependency-ordered deployment, hash-based idempotency and OIDC authentication.
- Resolved stale Redis API responses by selectively invalidating affected keys within 5 minutes of Snowflake refreshes, using a Python Azure Function to poll processing logs when direct notifications were unavailable.
- Automated ETL failure and data quality notifications with Python Azure Functions checking every 15 minutes, integrating structured AppDynamics events and email alerts with daily refresh reports across test and production.
- Added Claude risk summaries to pricing comparisons using domain-context prompts to flag missing claim types and regional value changes, isolating model failures to preserve deterministic validation and report generation.
- Extended C# ASP.NET Core REST APIs and Angular dashboards used by 4,000+ internal users, adding product usage, service insights and customer value views with regional and customer hierarchy filters.

Software Engineering Co-op Aug 2024 - Aug 2025
Ecolab *St. Paul, MN*

- Developed three Snowflake ML recommendation strategies for a catalog of 1,000+ products, applying Apriori to purchase associations, KMeans to customer segmentation and SVD to customer similarity.
- Established repeatable pipeline validation with pytest suites covering business rules, null handling, schema structure and ingested data, supporting correctness checks for production pipeline changes.
- Improved partition pruning for Snowflake queries serving production APIs and dashboards through table clustering, SQL tuning and database object redesign when query-level changes were insufficient.
- Built the initial pricing validation tool with Python, Streamlit and Snowpark, identifying missing records and changed values by comparing source workbooks with Snowflake data before promotion across environments.

Software Engineer Oct 2020 - Jul 2022
Tata Consultancy Services *Mumbai, India*

- Validated trading workflows during a TIBCO-to-ActiveMQ migration, checking message processing, trade-specific business rules and list-based eligibility decisions to verify behavior across the messaging transition.
- Automated trading-platform regression checks with UFT, VBScript and PowerShell, replacing repeated manual checks with reusable scripts for release validation.

PROJECTS

LLM Hallucination Detection and Mitigation | Python, Sentence Transformers, scikit-learn

- Scored semantic uncertainty using sentence embeddings and agglomerative clustering, adding retrieval-grounded revisions, proposer/skeptic prompts and citation-or-abstention checks to flag potentially unsupported responses.
- Built evaluation scripts reporting detection precision, recall and F1 alongside before-and-after reference consistency, using a DeBERTa natural language inference model to assess contradictions and entailment.

Claim Verifier | Python, FastAPI, ASP.NET Core, PostgreSQL

- Preserved claim submissions during inference failures by integrating a FastAPI vision-analysis service with ASP.NET Core and PostgreSQL, separating model observations from deterministic claim decisions.

Custom TCP Protocol | JavaScript, Node.js, TCP

- Handled fragmented and coalesced TCP messages with a buffered binary-protocol parser, decoding length-prefixed headers and retaining incomplete frames until enough bytes arrived to process each message.

TECHNICAL SKILLS

Languages: Python, SQL, C#, TypeScript, JavaScript
Frameworks and Libraries: PyTorch, pandas, NumPy, scikit-learn, FastAPI, ASP.NET Core, React, Angular, Streamlit
Databases: Snowflake, PostgreSQL, MySQL, MongoDB, Redis
Cloud Platforms: Azure (FunctionApps, OIDC, Key Vaults, Blob), AWS (EC2, EKS, ECR, DynamoDB, Bedrock, Sagemaker)
Tools: Docker, Kubernetes, GitHub Actions, Git, pytest, Playwright, AppDynamics

EDUCATION

Rochester Institute of Technology Dec 2025
Master of Science in Computer Science